

MD ASIF HASAN

Dhaka, Bangladesh

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Education

B.Sc. in Computer Science and Engineering
East West University, Dhaka, Bangladesh
CGPA: 3.08/4.00

Graduation: June 2024

M.Sc. in Computer Science and Engineering
East West University, Dhaka, Bangladesh
CGPA: 3.65/4.00

February 2025 – Present

Technical Skills

Programming Languages: Python, C, C++, Java, JavaScript, PHP.

Machine Learning & AI: TensorFlow, PyTorch, Scikit-Learn, CUDA, OpenCV.

Proficient in developing, training, and deploying machine learning models for image processing, time series forecasting, and classification. Skilled in deep learning architectures (CNNs, RNNs, LSTMs, Transformers), reinforcement learning (PPO, DQN, DDPG), and self-supervised learning (SimCLR, MoCo, BYOL). Experienced in feature engineering, model optimization, and GPU-accelerated training.

Blockchain & Cryptography: Blockchain development, consensus algorithms, post-quantum cryptography.

Developed a custom cryptocurrency (ASICoin) integrating post-quantum cryptography. Designed novel blockchain consensus mechanisms incorporating computational power, uptime, storage, token burning, and reputation scoring. Experience in blockchain security, smart contracts, and AI-based fraud detection.

Full Stack Development: MERN Stack (MongoDB, Express, React, Node.js), Flask, RESTful APIs.

Experienced in building responsive, scalable web applications with secure backend services, API integration, and modern UI/UX practices.

Data Science & Analytics: Pandas, NumPy, Matplotlib, Seaborn, Plotly, SQL.

Strong knowledge of exploratory data analysis (EDA), statistical modeling, correlation analysis, and visualization for actionable insights.

Embedded Systems & IoT: Arduino, Raspberry Pi, IoT Protocols.

Hands-on experience with embedded systems development, sensor integration, and real-time data processing for IoT applications.

Networking & Security: Delay-Tolerant Networking (DTN) protocols, HAM radio communications.

Practical experience in designing resilient communication systems for challenging environments.

Cloud & DevOps: Docker, Git, Linux.

Familiar with containerization, cloud deployment, and version control for collaborative development workflows.

Research & Publications

Conference Papers:

- **Ensemble Deep Learning for Enhanced Detection and Classification of Hand, Foot, and Mouth Disease**
International Conference on Data Mining and Information Security 2024.
- **Classification of Peptic Ulcer and Digestive Tract Disorders Using Endoscopic Images**
International Conference on Data Mining and Information Security 2024.
- **Rose Disease Detection Using YOLO: A Deep Learning-Based Approach**
9th International Conference on ICT for Competitive Strategies.

Research Experience

- **The Opportunistic Network Environment (ONE) Simulator**
Role: Team Lead — Tech: Java, Python
Led the development of a mobile application incorporating Delay Tolerant Network (DTN) routing protocols for real-world simulations. Conducted in-depth analysis of DTN protocols, focusing on optimizing data routing and delivery in environments with intermittent or unreliable network connectivity. Integrated machine learning algorithms to enhance routing efficiency and adaptiveness, thereby improving the overall performance of the network in varying real-world conditions. Collaborated with a multidisciplinary team to bring this project to fruition, applying the ONE simulator to a range of applications, including disaster relief and remote area communication systems.
- **Image Prediction using Self-Supervised Learning**
Role: Team Lead — Tech: Python, PyTorch, CUDA
Led the design and implementation of an image classification system leveraging self-supervised learning techniques to reduce reliance on labeled data. Experimented with advanced frameworks such as SimCLR, MoCo, and BYOL, integrating them with ResNet-based backbones for robust feature extraction. Preprocessed and augmented large-scale datasets to enhance representation learning and generalization. Utilized GPU acceleration with CUDA to optimize training efficiency and scalability. Collaborated with team members to benchmark performance against supervised baselines, achieving significant accuracy improvements in low-label scenarios, with applications in domains such as medical imaging and wildlife monitoring.
- **Sentiment Analysis for Stock Market Prediction**
Role: Lead Developer — Tech: Python, NLP, TensorFlow, LSTM, Sentiment Analysis
Developed an AI-powered stock market prediction model integrating sentiment analysis of financial news and social media data. Implemented NLP techniques, including BERT and VADER, to extract sentiment polarity from news articles and tweets. Combined sentiment scores with historical stock price data to train an LSTM-based predictive model. Optimized feature engineering using moving averages, volatility indices, and macroeconomic indicators to enhance prediction accuracy. Designed a Flask-based web interface to visualize sentiment trends and forecasted stock movements.
- **Hybrid News Recommender System**
Role: Lead Developer — Tech: Python, Scikit-Learn, TensorFlow
Spearheaded the development of a hybrid news recommender system by combining content-based filtering, collaborative filtering, and deep learning techniques. Leveraged machine learning models to personalize news recommendations based on user preferences and historical data. The system incorporated TensorFlow to train deep neural networks and Scikit-Learn for data preprocessing and model evaluation. Performance was rigorously tested using Microsoft's MIND dataset, yielding promising results in terms of user engagement and recommendation accuracy. The project aimed at enhancing the user experience on news platforms by providing tailored content based on both content attributes and collaborative user behavior.

- **Cryptocurrency Price Prediction Using LSTM-PPO**

Role: Lead Developer — Tech: Python, PyTorch, LSTM, PPO

Designed and implemented an LSTM-based deep learning framework to forecast Ethereum (ETH) prices using historical market data. Incorporated multiple technical indicators—including moving averages, RSI, and MACD—to enhance predictive accuracy. Performed sentiment analysis on cryptocurrency-related news to capture market mood and integrate it into the forecasting pipeline. Developed a deep reinforcement learning module using Proximal Policy Optimization (PPO) to execute trades based on model predictions, enabling adaptive decision-making in dynamic market conditions. Achieved strong performance in both short-term and long-term forecasting, providing actionable, data-driven insights for trading strategies.

Extracurricular Activities

- **Scouting:** Senior Rover Mate, Messenger of Peace Ring Badge recipient, National Service Award for outstanding community work. Engaged in leadership roles, organized social outreach programs, and mentored younger scouts.
- **Debating:** Former Trainer and university debating club General Secretary. Led training sessions, coordinated events, and contributed to the university-level debating team becoming champions in 2023. Represented the university in several national debate competitions.
- **Competitive Programming:** Actively solved thousands of algorithmic problems across platforms such as Codeforces, Hackerrank, and Beecrowd. Developed strong problem-solving and data structure skills, achieving top rankings in several contests. A regular participant in online and offline hackathons.
- **Leo Club Membership:** Served as Treasurer (2019-2020) of the Leo Club of Dhaka Imperial. Managed finances for events, fundraisers, and charitable activities. Organized community service projects such as blood donation drives and environmental awareness campaigns.
- **Martial Arts:** Achieved Blue Belt in Karate and Green Belt in Taekwondo. Actively participated in local tournaments and practice sessions, demonstrating discipline, focus, and physical endurance. Promoted the values of martial arts in community activities.
- **Marathon Running:** Completed over 20 marathons and regularly participated in running events organized by Run Bangladesh. Contributed to event management and logistics for races, including four marathons as an event organizer. Helped promote healthy lifestyles through running campaigns.
- **HAM Radio Enthusiast:** Holder of a BTRC license and call sign S21NEO. Passionate about building and testing radio equipment, focusing on networking and antenna design. Actively involved in the HAM radio community, participating in national and international radio events and disaster relief communications.